Tuğcan Hoşer

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PROFILE

MSc student in Informatics specializing in **Machine Learning**, with **3+ years of experience** in **deep learning**, **multi-modal models (vision-language), model optimization**, and **deploying AI systems** to production. Seeking roles where I can contribute to building scalable, real-world AI products.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer *Neuracast-ai*
Mar 2025 – Present
Munich, Germany

- Designed a **linear programming-based allocation engine** integrating demand forecasts, capacity constraints, and campaign priorities.
- Implemented **adaptive anomaly detection** to handle unexpected demand shifts and improve system stability.
- Achieved up to **95% variance accuracy** when compared to real-life allocation scenarios, validating model performance on live retail data.
- Built a modular backend with real-time stock update handling and **interactive visualizations** for allocation transparency.

Machine Learning Researcher
Munich Technical University
Mar 2025– Jul2025
Munich, Germany

- Conducting research on force field modeling using Graph Neural Networks.
- Designing models to accurately predict energy and forces of molecules and atoms, especially in **long-range force fields**.
- Working with graph-structured data and semantic representations to improve scientific modeling.
- Collaborating with academic researchers to prototype scalable ML solutions.

Software Developer Intern
Oncosia Scientific GmbH
Nov 2024 – Mar 2025
Munich, Germany

- Developed and maintained **backend services** using **Django**, ensuring scalability and performance.
- Managed **DevOps workflows** with **Docker, Ansible, and Linux**, integrating **CI/CD pipelines** for automated deployments.
- Optimized **cloud-based** applications, improving efficiency for **large-scale AI models**

Deep Learning Intern
PreciTaste
Sep 2024 – Feb 2025
Munich, Germany

- Developed and optimized **LLM-powered applications** for multimodal AI and data querying
- **Optimized inference speed and memory usage**, implementing **image query embedding caching** to reduce computational costs.
- Researched and implemented **LLM-powered solutions** for **AI-driven** applications.
- Applied model **quantization** to achieve faster, more efficient inference in real-world deployment settings.

ML/DL Research Internship
TUM Klinikum rechts der Isar
May 2024 – Aug 2024
Munich, Germany

- Tested **ProtoASNet, ProtoPFormer, and PiPNet** for medical image classification, focusing on accuracy and interpretability.
- Created and analyzed **heatmaps** to make model decisions easier to understand, using feedback from users.
- Improved **deep learning models** for medical use by adding uncertainty measures to make decisions more reliable.

Full Stack Developer Intern
Koç Finansman A.Ş
Jul 2021 – Aug 2022
Istanbul, Turkey

- Developed and deployed **full-stack applications** using **Spring Boot, React.js, and Apache Spark** in an **Agile** environment.
- Worked with **Docker, Git, and Linux** to manage deployments and optimize system performance.
- Collaborated with senior developers to **improve backend efficiency** and integrate **DevOps practices**.

EDUCATION

[MSc] Computer Science — Apr 2023 - present
Munich Technical University — Munich, Germany
- **Major:** Machine Learning
- **Minors:** Computer Vision & Software Engineering
- Completed coursework in Introduction to Deep Learning, Machine Learning, Advanced Deep Generative Models, AI in Medicine, 3D Geometry, and Computer vision courses.

[BSc] Computer Engineering — Sep 2017 - Jul 2022
Ozyegin University — Istanbul, Turkey
- **GPA:** 3.45/4.00
- **Teaching Assistant** for **CS 101** course
- **Research Assistant** at **Emre Sefer Lab**

PROJECTS

Interpretable AI for Medical Classification using ProtoPNet concept — Apr 2024 - Aug 2024
- Research

Stock Price and Direction Prediction via Deep Attention-Based CNN — Sep 2021 - Jun 2022
- Applied **ConvMixer and Vision Transformers** to financial data by converting time-series into **2D image representations**.
- Evaluated **self-attention and patch embeddings** for improving **forecasting performance**.
- Presented findings at **ICAIF 2022**, showcasing deep learning's role in financial predictions. Research

PROGRAMMING LANGUAGES

Python	●●●●●	C++	●●●●
Java	●●●●	Javascript	●●●●
C	●●●		

LIBRARIES - FRAMEWORKS

Machine Learning - Deep Learning
- Pytorch • Ollama
- Tensorflow / Keras
- Weights & Biases
- Huggingface
- Numpy
- Pandas
- Scikit-learrn
- Matplotlib
- Langchain
- CUDA
- OpenCV

Other Knowledge
- Docker • Git • Ansible
- Linux (Ubuntu)
- CPU/GPU
- RAG
- Latex
- ReactJS
- Django
- Spring Framework
- Google Cloud
- AWS
- Virtual Machine
- SQL

INTERESTS

- Machine Learning
- Multi Modal Learning
- Large Language Models
- High Performance Computing
- AI Model Optimization
- Deep Learning
- Generative AI
- Object Detection / Segmentation
- Transformer based models
- Competitive Programming